

Benchmark paper

Anonymized Authors

Anonymized Affiliations
email@anonymized.com

Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility.

We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models.

It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. Standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI.

1 Introduction (Reuben)

First paragraph:

- Context: Diffusion MRI provides quantitative microstructural biomarkers
- Increasing use of ML for prediction (age, diagnosis, cognition, etc.)
- Lack of standardized benchmark

Second paragraph: literature review:

- Literature review of techniques: Variability of the proposed approach (ML-based, deep-learning based etc)
- Problems: High variability in reported performance across studies + Limited reproducibility

Third paragraph: reasons for these problems

- Reason 1: Heterogeneous preprocessing pipelines
- Reason 2: Inconsistent feature representations (voxel-wise vs ROI-based), not analysis cross microstructure features

- Reason 3: Evaluation protocols not directly comparable + results on different datasets
- Reason 4: Missing code
- Our solution: "To address these challenges, we propose: 1) an open-source and standardized benchmark; 2) of various machine learning techniques in dMRI; 3) on several tasks (classification, regression) 4) encompassing N datasets

Fourth paragraph: your contributions

- Standardized preprocessing pipeline for diffusion MRI
- Implemented various representation of the microstructural input data
- Unified feature extraction strategy
- Controlled evaluation protocol (train/test splits)
- Comparison of multiple feature representations
- Integration of both classical and DL predictors
- Public release of code and benchmark configuration
- End-to-end analysis

2 Benchmark (Reuben)

Start with a general introduction of this section that introduce each component of your benchmark + add a figure that presents an overview.

2.1 Datasets

- Introduction sentence about why you picked these datasets
- Introduce Number of subjects
- Age range / demographics
- Scanner details
- Acquisition protocol or summary table with acquisition parameters
- Prediction targets (classification/regression)

If space (unlikely): * PLOTS: Demographic distribution (age histograms) | Table with demographics summary (gender classes, age range...)

2.2 Unified diffusion data preprocessing

Data preprocessing Following..., we propose to

- Denoising
- Motion and eddy correction
- Spatial normalization
- Masking strategy

Microstructural Representation Explain here the different forms of input representation. The word "feature representation" is ambiguous because neural network also extract features in a data-driven way. That's why I would rather use the term like "Microstructural Representation". Explain why differences between white and gray matter. Give references here about why there are differences.

Gray Matter Representation

- Voxel-wise features
- High-dimensional representation (64k features)
- Mask-based extraction

Explain that voxel-wise features preserve spatial resolution but increase dimensionality and risk of overfitting.

White Matter Representation

- Tract-based aggregation
- Mean values per tract (48 features)
- Reduced dimensionality
- Anatomical interpretability

Justify tract-averaged representation: Noise reduction, improved stability, lower dimensional feature space, anatomically meaningful structure.

Microstructural normalization

- Binary targets (encoded 0/1), Continuous targets (standardized during training)
- Standardization fitted on training set only
- Applied to validation/test sets
- Inverse transform for regression outputs (for visualization)

2.3 Prediction models

- Dummy baseline (e.g., majority class for classification, mean predictor for regression)
- Classical ML pipeline: Linear models (e.g., logistic regression, linear regression), Random Forests, Support Vector Machines (SVM) + their PCA versions
- Deep Learning models (e.g., dinov2, medicalnet)
- Unified hyperparameter search strategy (e.g., grid search or random search with cross-validation on the training set)

2.4 Benchmarks tasks and evaluation metrics

Task selection rationale: Clinical relevance, data availability, and established benchmarks in the literature.

Classification

- Task description: Binary diagnosis prediction (e.g., ASD vs. TD)
- Evaluation Metrics: Accuracy, AUC, F1-score

Regression

- Task description: Age prediction (brain age estimation)
- Evaluation Metrics: R^2 , MAE

2.5 Evaluation protocols

- Data Splitting Strategy (Fixed train/test splits)
- Cross-validation within training set. Stratification
- No information leakage
- Mean $\pm$ std across folds
- Statistical comparison tests (e.g., paired t-tests, Wilcoxon signed-rank tests) to assess significance of performance differences between models

3 Results (Reuben)

Introduction sentence

Choice of prediction models Table: For all tasks, choose one brain structure and one microstructural representation (choose widely this one) and report results in a table + comment

Choice of the brain structure For one regression task and one classification task, show the impact of the brain structure with one microstructural representation. You can discard models that underperformed in the first model section.

Choice of the microstructural representation For one task, show the impact of the microstructural representation for each brain structure. You can discard models that underperformed in the first model section.

Training size impact I think we will not have space for this, but here it would be for one regression task and one classification task, one structure, one microstructural representation, impact of the model, here try to have variability (PCA, non-PCA, deep) for example.

Methodological insights

- DL vs ML based
- Importance of dimensionality control
- Effect of tract-based aggregation
- Risk of overfitting in mesh representations without PCA bcs of the high number of vertices representations

4 Conclusion (Reuben)

- Introduced standardized dMRI benchmark for age prediction in healthy individuals using three large datasets (HCP, CamCAN, ABIDE)
- Compared feature representations (MD, SH, MK, RTOP) and models (dummy classifier, linear, pca_linear, forest, pca_forest, svm, pca_svm)
- Provided reproducible end-to-end framework and enables fair model comparison. Reduces methodological heterogeneity
- A step toward methodological unification/code availability

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These inconsistencies largely stem from four recurrent factors. First, studies often rely on heterogeneous preprocessing pipelines (and sometimes different quality-control criteria), making results difficult to compare across cohorts. Second, diffusion information is represented in markedly different ways (e.g., voxel/vertex-wise maps, tract/ROI summaries, or learned embeddings), and evaluations rarely disentangle how the choice of microstructural representation affects downstream performance. Third, evaluation protocols vary substantially (e.g., different split strategies, cross-validation schemes, and hyperparameter tuning procedures), and models are frequently assessed on different datasets, further confounding comparisons. Finally, incomplete release of code, configurations, and split definitions limits reproducibility and slows iterative progress. To address these challenges, we propose an open-source and standardized benchmark that unifies preprocessing and feature extraction, evaluates a diverse set of machine learning approaches on both regression and classification tasks, and spans multiple publicly available datasets under a controlled and leakage-free evaluation protocol.

In this work, we introduce an open-source, end-to-end benchmark for predictive modeling with diffusion MRI that targets reproducibility and fair comparison across studies. Our contributions are: (i) a standardized preprocessing and data preparation pipeline; (ii) a unified feature-extraction framework supporting multiple microstructural representations for gray and white matter; (iii) a controlled evaluation protocol with consistent, leakage-free train/validation/test splits and cross-validation; (iv) a systematic comparison of feature representations and predictors, spanning classical machine learning baselines and deep-learning models; and (v) the public release of code, configurations, and scripts to facilitate transparent reuse and extension.

6 Benchmark

Start with a general introduction of this section that introduce each component of your benchmark + add a figure that presents an overview.

Our proposed benchmark provides a standardized, end-to-end framework for evaluating predictive models on diffusion MRI data. As illustrated in Figure 1, the pipeline comprises several interconnected modules: uniform preprocessing, microstructural feature extraction, predictive modeling, and rigorous evaluation. Specifically, our benchmark applies a unified preprocessing pipeline to all cohorts—with the exception of the HCP dataset, which utilizes its own established preprocessing pipeline. Following preprocessing, we perform tailored feature extraction to capture tissue-specific microstructural representations. Finally, we apply a rigorous, standardized cross-validation and analysis protocol to evaluate a wide spectrum of predictive models, ensuring reproducibility and fair comparison.

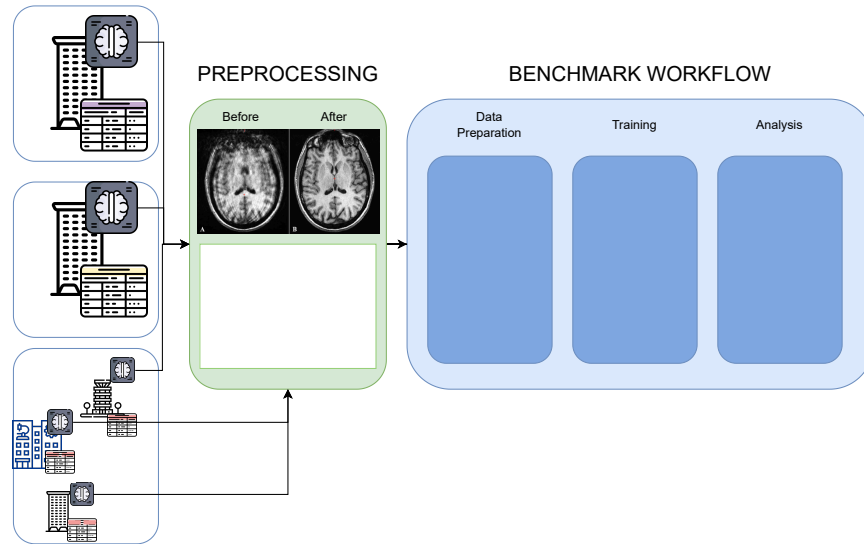


Fig. 1. Overview of the proposed diffusion MRI benchmark pipeline, illustrating the end-to-end process from preprocessing to microstructural feature extraction, predictive modeling, and evaluation.

6.1 Datasets

To ensure reproducibility, we evaluate our benchmark on three publicly available datasets covering both healthy and clinical populations, offering diverse demographics, scanner types, and acquisition protocols. For healthy cohorts, we utilize the Human Connectome Project (HCP) Young Adult dataset and the Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset, both targeting gender and age prediction (classification and regression, respectively).

After filtering, the HCP cohort comprises 1,036 subjects (474 males, 562 females; ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b -values = 1000, 2000, 3000 s/mm², 90 directions/shell). The Cam-CAN cohort includes 629 subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b -values = 1000, 2000 s/mm², 30 directions each).

For clinical evaluation, we use the Autism Brain Imaging Data Exchange II (ABIDE II) dataset, chosen over ABIDE I due to the availability of diffusion MRI. Targeting Autism Spectrum Disorder (ASD) diagnosis (classification), the filtered ABIDE II cohort consists of 218 subjects (186 males, 32 females; ages 5-64), including 119 with ASD and 99 controls. As a multi-site dataset, ABIDE II utilizes various 3T scanners (e.g., Siemens, GE) with highly heterogeneous, site-specific single-shell diffusion protocols. Across the included sites, spatial resolutions range from 0.94 to 3.0 mm, TR/TE from 5200-20244/78-101 ms, b -values include 1000, 1500, and 2500 s/mm², and diffusion directions vary between 32 and 64. This extreme variability provides a realistic scenario for evaluating model robustness to scanner heterogeneity.

6.2 Preprocessing and Feature Extraction

Preprocessing (Himanshu)

- Denoising
- Motion and eddy correction
- Spatial normalization
- Masking strategy

Microstructural Representation

Gray Matter Representation Gray matter microstructure was represented on the subject-specific cortical surface reconstructed from **T1-weighted images using the Desikan–Killiany atlas (aparc)**. For each subject, cortical meshes (white/pial surfaces) were generated and diffusion-derived scalar maps (e.g., MD, SH-based metrics) were computed in native diffusion space and rigidly aligned to the anatomical space. Voxelwise diffusion measures were then projected onto the individual cortical surface using ribbon-constrained sampling, preserving vertex-wise resolution (64k vertices per hemisphere). This subject-specific surface representation was chosen to respect cortical geometry, reduce partial-volume effects, and maintain high spatial specificity while enabling anatomically consistent regional correspondence via the atlas labels across subjects.

White Matter Representation White matter microstructure was represented within a standardized tract space using the Tract-Based Spatial Statistics (TBSS) framework. Diffusion-derived scalar maps were nonlinearly registered to standard

space, projected onto the group-wise white matter skeleton to minimize partial volume effects and residual misalignment, and summarized using the JHU White-Matter Tractography Atlas. Mean microstructural values were extracted from 48 major bundles defined in the atlas, yielding a compact tract-level representation per metric. We intentionally adopted this skeleton-based, atlas-constrained strategy rather than subject-specific tractography to enforce anatomical correspondence across subjects, reduce variability introduced by tracking parameters, and enhance robustness in a multi-dataset benchmarking setting. Although this representation sacrifices voxel-level granularity relative to the cortical surface analysis, it provides reproducible, homologous bundle definitions across datasets and mitigates registration and tract delineation inconsistencies—thereby strengthening cross-cohort comparability and methodological stability.

Microstructural models and normalization To comprehensively capture the biological properties of brain tissue, we compute a diverse set of microstructural metrics using established diffusion models. These metrics were selected to represent different facets of tissue architecture, ranging from basic diffusion magnitude to complex restricted environments:

- **Baseline ($b = 0$):** The averaged T2-weighted non-diffusion signal is included as a baseline to disentangle purely diffusion-driven predictive power from underlying structural T2 contrast.
- **Mean Diffusivity (MD):** Computed via Diffusion Tensor Imaging (DTI), representing the overall magnitude of water diffusion.
- **Mean Kurtosis (MK):** Derived from Diffusion Kurtosis Imaging (DKI), capturing the non-Gaussianity of diffusion as a proxy for microstructural heterogeneity and tissue complexity.
- **Return-To-Origin Probability (RTOP):** Estimated using a Laplacian-regularized Mean Apparent Propagator (MAP-MRI) model. RTOP reflects the probability of water molecules remaining in their initial position, serving as a strong indicator of cellular restriction and density.
- **Spherical Harmonics (SH) Power:** A model-free representation obtained by computing the L2 norm of the SH coefficients (up to order 6) directly from the diffusion signal, capturing signal complexity and anisotropy without assuming a specific tissue compartment model.

We apply a robust ventricular normalization strategy. For each subject, voxel-wise diffusion metrics are divided by the mean metric value extracted from the cerebrospinal fluid (ventricles). Finally, extreme pathological values are clipped at the 99th percentile to ensure numerical stability during model training.

6.3 Prediction models

To establish a comprehensive evaluation, our benchmark incorporates a wide spectrum of predictive models, ranging from simple baselines to advanced deep learning architectures. First, we include dummy estimators (majority class for

classification and mean predictor for regression) to provide a lower-bound performance baseline. For classical machine learning, we evaluate linear models (e.g., logistic regression, Lasso), Random Forests, and Support Vector Machines (SVM). Given the high dimensionality of the microstructural representations, we also evaluate these models coupled with Principal Component Analysis (PCA) for feature reduction.

For deep learning, we benchmark several vision architectures. We leverage pre-trained foundation models, specifically DINOv2 and Curia, utilizing them as frozen feature extractors where representations are computed offline prior to training the prediction head. In contrast, 3D architectures like MedicalNet are fully fine-tuned on our diffusion datasets to adapt to the specific microstructural domains. **To ensure a fair and rigorous comparison, all models are optimized using a unified hyperparameter search strategy (e.g., grid or random search) with internal cross-validation strictly confined to the training set to prevent data leakage.**

6.4 Benchmarks tasks and evaluation metrics

The selection of prediction tasks in our benchmark is driven by clinical relevance, data availability, and the need to align with established benchmarks in the neuroimaging literature. We evaluate models across two primary paradigms:

Classification For binary classification, we focus on tasks such as Autism Spectrum Disorder (ASD) diagnosis (ASD vs. Typically Developing) using the ABIDE II dataset, as well as gender prediction in the HCP and Cam-CAN cohorts. For all classification tasks, binary targets are strictly encoded as 0 and 1. To comprehensively assess classification performance, especially in the presence of class imbalances, we compute standard and weighted accuracy, precision, recall, and the F1-score.

Regression For continuous targets, we evaluate brain age estimation using the healthy cohorts (HCP and Cam-CAN). During the training of Deep Learning models, these continuous targets are standardized. To prevent data leakage, the standardization parameters are fitted exclusively on the training set and subsequently applied to the validation and test sets. Model performance is quantified using a robust set of metrics to capture different aspects of the error distribution: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) (both standard and sample-weighted), R^2 score, explained variance, Mean Absolute Percentage Error (MAPE), and the Pearson correlation coefficient.

6.5 Evaluation protocols

To rigorously evaluate the predictive models, we employ a stratified k -fold cross-validation strategy. This ensures fixed, reproducible train and test splits while

maintaining the **original target distribution (e.g., class balance for classification or target range for regression)** across all folds. Strict separation between training and testing data is maintained at all stages; preprocessing steps, feature normalization, and hyperparameter tuning are fitted exclusively on the training set of each fold and subsequently applied to the held-out test set, preventing any information leakage. Model performance is aggregated across all folds and reported as the mean $\pm$ standard deviation for each evaluation metric.

7 Results

7.1 Within-Dataset Performance

- Model/Feature/Tissue comparison
- Mean $\pm$ std of best run / model
- Statistical testing
- PLOT: Bar plots per dataset | Camcan-HCP comparison(?)

7.2 Feature Representation Comparison

- Gray vs white matter
- Microstructure metrics comparison
- PLOT: Tissue performance according to microstructure metric for task purposes | Microstructure features

7.3 "Handmade" vs DL embeddings

7.4 Training Size Analysis

* PLOT: Learning Curve

7.5 Key Findings

- Which models perform best? Which feature representations perform best? Tissue predominance in task prediction?
- Are simpler models sufficient? Are DL models necessary for predicting demographics?
- Stability across folds?

7.6 Methodological Insights

- Importance of dimensionality control
- Effect of tract-based aggregation
- Risk of overfitting in mesh representations without PCA bcs of the high number of vertices representations

8 Conclusion

- Introduced standardized dMRI benchmark for age prediction in healthy individuals using three large datasets (HCP, CamCAN, ABIDE)
- Compared feature representations (MD, SH, MK, RTOP) and models (dummy classifier, linear, pca_linear, forest, pca_forest, svm, pca_svm)
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8.1 Limitations/Further steps

- No multi-site harmonization (explicitly as scope limitation). Would training in one dataset and predicting in another dataset have the same effect as predicting in different folds of the same dataset? Are the predictions very far?
- Limited tasks / ranges in prediction (e.g. HCP age prediction)
- Limited preprocessing / including other preprocessing parameters or corrections

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. Multiple citations are grouped [?], [?].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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